# Synthetic Cold Splits Are Not Enough: Distribution-Shift Benchmarking for Drug-Target Interaction Prediction

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**Abstract**

Drug-target interaction (DTI) papers often use synthetic cold-start splits as proxies for generalization, but it remains unclear whether conclusions drawn from these splits transfer to real distribution shifts. We present a benchmark-first case study showing that, within an expanded five-model panel, rankings are highly sensitive to how the shift is defined. The panel contains a fully specified fixed-feature `base` encoder, a learner-control `RandomForest` baseline on the same Morgan+k-mer inputs, two recent pooled-LM baselines (`DTI-LM` and `HyperPCM`), and a `GraphDTA-style` graph-family probe. On a 3-seed synthetic reference panel built from `BindingDB_Kd`, the winner depends strongly on the split: `RF` is strongest on `unseen_drug` ($0.7339 \pm 0.0051$ AUPRC) and `blind_start` ($0.4512 \pm 0.0373$), while `base` remains strongest on `unseen_target` ($0.5224 \pm 0.0602$). On the promoted temporal/provenance benchmarks, the ranking changes again. Over 3 seeds, `RF` becomes strongest on both `BindingDB_patent / patent_temporal` ($0.8234 \pm 0.0012$) and `BindingDB_nonpatent_Kd / nonpatent_temporal` ($0.7201 \pm 0.0012$), with paired bootstrap intervals remaining strictly positive against both `base` and `DTI-LM`. In contrast, the `GraphDTA-style` probe is competitive but does not dominate any promoted axis. These additions strengthen the benchmark claim rather than weakening it: ranking instability is not confined to pooled-LM comparisons, and even a classical learner on fixed handcrafted inputs can become the strongest system under both synthetic and temporal evaluation. A pooled year-band and overlap-bucket diagnosis further shows that temporal/provenance OOD is structured rather than monolithic. We package the split manifests, benchmark resources, and standalone evaluator for reuse. Within the studied model panel and benchmarks, these results show that synthetic cold splits and real OOD shifts should not be treated as interchangeable evaluation settings for DTI.

## 1 Introduction

Predicting drug-target interactions for unseen molecules or proteins is a central goal in computational drug discovery. Because truly prospective evaluation is difficult and expensive, many recent studies report held-out drug, target, or blind splits on datasets derived from BindingDB or DAVIS as proxies for inductive generalization [1, 2, 3, 4, 5, 6, 7, 8]. These protocols are useful, but they also encourage a strong implicit assumption: if a method performs well on one cold-start split, then it is broadly robust to the kinds of shift encountered in deployment.

That assumption is weaker than it appears. Different split definitions remove different forms of overlap, and they may favor different model families. A pooled-LM baseline that looks strong under one synthetic exclusion rule may no longer be the most reliable option once the benchmark shifts in time, provenance, or assay composition. If these ranking changes occur even within a compact studied panel, benchmark design is not a technical footnote. It becomes part of the scientific claim.

This paper studies that question directly. Rather than proposing another universal architecture, we ask whether the ranking of an expanded DTI model panel is preserved across synthetic cold-start regimes and real distribution shifts. The goal is to test ranking stability under benchmark shift in a controlled case

study, not to declare a universal field-wide leaderboard. Our panel begins with three representative systems: a fixed-feature MLP reference model (`base`) and two recent external literature baselines, `DTI-LM` and `HyperPCM`. We then add two deliberately different controls: a `RandomForest` classifier trained on the same Morgan+k-mer representation as `base`, and a `GraphDTA-style` graph-family baseline. Even on the synthetic reference panel, rankings are already heterogeneous: `RF` is strongest on synthetic `unseen_drug` and `blind_start`, while `base` remains strongest on synthetic `unseen_target`.

We then contrast this synthetic picture with real temporal/provenance benchmarks built from BindingDB patent and non-patent time-based splits. The result is another ranking change. The synthetic panel does not point to a single dominant system, but the promoted temporal benchmarks strongly favor `RF`. Paired bootstrap confidence intervals support these temporal comparisons: `RF - base` and `RF - DTI-LM` remain positive on both `patent_temporal` and `nonpatent_temporal`. An earlier-cutoff variant of the patent benchmark (`patent_temporal_v2017`) had already preserved the same temporal ordering within the smaller panel, so the expanded results strengthen rather than overturn the original benchmark diagnosis.

To move beyond leaderboard-style reporting, we also diagnose the promoted temporal benchmark by year and overlap bucket. This analysis shows that temporal OOD is not simply "harder cold-start." Instead, the gap is concentrated in specific year bands and overlap regimes, which helps explain why a method that looks strong under one split can lose under another. We further include matched 3-seed support on `BindingDB_Ki` and `DAVIS` to test whether the same external panel remains stable across shifted target-cold settings. Recent molecular benchmark studies have made analogous arguments about OOD sensitivity in adjacent tasks [9, 10, 11]; our contribution is to show the same issue concretely in DTI and to release the benchmark suite as a reusable split-plus-evaluator resource for reliable DTI evaluation.

Our contributions are fivefold. First, we construct a 3-seed synthetic reference panel showing that even standard cold-start evaluation is regime-dependent. Second, we establish real temporal/provenance benchmarks on patent and non-patent BindingDB slices and show that their ranking can differ sharply from the synthetic panel. Third, we expand the model panel beyond pooled-LM baselines with a learner-control `RF` system and a graph-family `GraphDTA-style` probe, showing that benchmark sensitivity is not confined to one architecture family. Fourth, we provide year-band and overlap-bucket analyses that connect the ranking change to measurable shift axes, matched 3-seed support on `BindingDB_Ki` and `DAVIS`, and two review-driven robustness probes. Fifth, we release canonical split manifests, reusable benchmark resources, and a standalone evaluator to make the suite easy to reuse.

## 2 Related Work

**Synthetic cold-start evaluation in DTI.** Cold-start evaluation has become standard practice in DTI and drug-target affinity prediction. Recent models routinely report random, held-out drug, held-out target, or blind-start splits on BindingDB-derived and DAVIS-derived benchmarks [3, 4, 5, 6, 7, 8]. This literature established synthetic cold-start testing as a useful proxy for inductive generalization. However, it often treats different split definitions as variations of the same evaluation problem. Our results suggest that this assumption is too coarse, because different synthetic regimes already produce different model rankings.

**Retrieval, inductive, and memory-based DTI models.** Another line of work designs architectures intended to transfer information from seen entities to unseen drugs or targets. Retrieval-based or transfer-oriented models attempt to reuse evidence from related compounds and proteins, while inductive or prototype-like systems aim to improve robustness under entity holdout [4, 5, 6, 7, 8]. Our work does not claim that these mechanisms are ineffective. On the contrary, we recover setting-specific gains for retrieval on synthetic

**Table 1:** Benchmark taxonomy and seed0 split statistics. Test prevalence differs substantially across benchmark families, so AUPRC should be compared within benchmark rather than across rows.

| Benchmark | Shift family | Synth.? | Test rows | Pos. rate |
|---|---|---|---|---|
| `Kd / unseen_drug` | drug-cold | yes | 12,744 | 0.1966 |
| `Kd / blind_start` | drug+target-cold | yes | 2,821 | 0.2031 |
| `Kd / unseen_target` | target-cold | yes | 12,052 | 0.2239 |
| `patent / patent_temporal` | temporal+source | no | 49,028 | 0.5745 |
| `Ki / unseen_target` | assay+target-cold | mixed | 70,380 | 0.4652 |
| `DAVIS / unseen_target` | dataset+target-cold | mixed | 5,168 | 0.0615 |

`blind_start` and for target-memory on synthetic `unseen_target`. The central point is different: benchmark choice determines whether these gains appear.

**Distribution shift and benchmark design.** Recent benchmark studies in molecular machine learning and related biomedical prediction problems have shown that model rankings can change under scaffold, temporal, or domain-shifted evaluation [9, 10, 11]. Our paper brings that diagnostic perspective into DTI. Unlike a leaderboard paper, we intentionally keep the model panel compact. The main result is not that one architecture dominates many baselines, but that even a small, representative panel is enough to expose ranking instability across benchmark definitions.

**Positioning.** Prior DTI papers typically optimize for one split regime or propose a stronger model within a fixed benchmark setup. This paper instead studies whether model rankings remain stable across synthetic cold-start regimes and a real-OOD patent benchmark, then diagnoses the shift structure behind disagreements.

## 3 Benchmark Setup

Our goal is to test whether an expanded DTI model panel preserves its ranking across synthetic cold-start regimes and real distribution shifts. The panel used in the revised manuscript contains five models. `base` is a fixed-feature MLP reference system built from RDKit Morgan fingerprints and hashed protein k-mer features. `RF` is a learner-control `RandomForest` classifier trained on the exact same handcrafted pair representation as `base`, allowing us to separate the effect of the learner from the effect of the representation. `DTI-LM` is a pooled ChemBERTa+ESM2 baseline aligned to recent literature, `HyperPCM` is a task-conditioned pooled-LM baseline, and `GraphDTA-style` is a graph-family probe with a molecular GCN and protein CNN. We intentionally keep the panel modest rather than exhaustive, but it now spans fixed-feature neural, fixed-feature classical, pooled-LM, and graph-based inductive biases. Because the anchor system is central to every ranking claim, Appendix Table A3 gives the exact `base` feature construction, architecture, parameter count, and training recipe.

We evaluate this panel on two benchmark groups. The first is a synthetic reference panel based on `BindingDB_Kd`, with three split definitions: `unseen_drug`, `blind_start`, and `unseen_target`. The second is a real-OOD screening panel. Our primary real-OOD benchmark is the BindingDB patent split (`patent_temporal`), which uses a year-based temporal partition and is evaluated over three seeds. We also include `BindingDB_Ki / unseen_target` and `DAVIS / unseen_target` as supporting shifted target-cold benchmarks that test whether the same external model family remains stable across assay and dataset shifts. Table 1 summarizes the shift taxonomy and split statistics.

We report AUPRC as the primary metric and AUROC as a secondary metric. AUPRC is emphasized because class balance and ranking behavior are central to the benchmark claims in this paper. For the expanded panel, we report matched three-seed results on the synthetic reference panel and the promoted temporal benchmarks. We first run transparent seed0 screening for the newly added baselines, then promote them where the screening result is either strong enough to affect the panel story or informative enough to test a reviewer concern. `RF` is promoted on all main and temporal axes because its seed0 performance is unexpectedly strong across the board. `GraphDTA-style` is promoted on `unseen_drug`, `unseen_target`, `patent_temporal`, and `nonpatent_temporal` because these are the axes most relevant to the benchmark claims and to the request for a graph-based family. Appendix Table A7 and the baseline-expansion report make this promotion logic explicit. We further add paired bootstrap uncertainty support for the headline expanded-panel comparisons, especially those involving `RF` on synthetic drug-cold, blind-start, and the temporal benchmarks. Appendix Table A6 collects these intervals.

To make the benchmark reusable, each split is exported as a canonical resource bundle with manifests, split tables, cached identifiers, and a standalone CSV-based evaluator. This resource layer allows external predictions to be scored against the exact same test partitions used in the paper. Appendix Table A8 enumerates the released artifacts and how they are consumed by the evaluator.

The patent benchmark requires several setup choices that are documented explicitly. We use a local year-based split of `BindingDB_patent`, with training years before the validation year, a held-out validation year, and a later temporal test set. Because this benchmark also changes provenance and assay composition, we describe it as temporal/provenance OOD rather than pure time shift. We also use a patent-specific affinity conversion rather than reusing the standard `pKd` transform, because the patent benchmark stores affinities in a different log-scale convention. Appendix Table A1 summarizes the exact year boundaries, split counts, label construction, overlap policy, and seed semantics for both `patent_temporal` and the alternative-cutoff `patent_temporal_v2017`. Appendix Table A2 provides the same information for the review-driven `BindingDB_nonpatent_Kd / nonpatent_temporal` probe. BindingDB-derived panels are anchored in the BindingDB resource [1], while DAVIS provides an external kinase-focused support benchmark [2].

Finally, for diagnostic analysis, we use the saved overlap score from each test prediction to compute overlap buckets that summarize how strongly each test example resembles the training support. Concretely, `overlap_score` is defined as $\min(s_d, s_t)$, where $s_d$ is the cosine similarity between the test drug's Morgan fingerprint and its nearest training-drug neighbor, and $s_t$ is the cosine similarity between the test target's hashed k-mer feature vector and its nearest training-target neighbor. We use the minimum to conservatively define overlap: a pair is high-overlap only when both its drug and target are well represented in the training set. We retain cosine for the released overlap score so that drug-side and target-side overlap share the same similarity scale inside this joint diagnostic; the separate synthetic similarity analysis still reports the standard Morgan-fingerprint Tanimoto and scaffold-overlap statistics. We report benchmark taxonomy and split statistics, including test-set size and prevalence, so that AUPRC is interpreted within benchmark rather than across benchmark families.

# 4 Results

## 4.1 Synthetic cold splits are already regime-dependent

We begin with the `BindingDB_Kd` synthetic reference panel. Table 2 shows the main pattern: even within a single dataset, model rankings are not stable across split definitions, and the ranking changes further once the learner-control and graph-family additions are made. On `unseen_drug`, the strongest model is no longer `HyperPCM` but `RF`, which reaches $0.7339 \pm 0.0051$ versus $0.5509 \pm 0.0900$ for `base`, $0.5694 \pm 0.0473$ for `HyperPCM`, $0.5004 \pm 0.0341$ for `DTI-LM`, and $0.5626 \pm 0.0172$ for `GraphDTA-style`. The paired

**Table 2:** Synthetic reference panel and promoted real-OOD benchmarks under the expanded five-model panel. Values are mean ± std over seeds. Best mean AUPRC in each row is bold. The classical learner-control baseline (`RF`) is trained on the same Morgan+k-mer inputs as `base`; `GraphDTA-style` provides the architecture-diverse graph-family probe.

| Benchmark | Seeds | base (AUPRC/AUROC) | RF (AUPRC/AUROC) | DTI-LM (AUPRC/AUROC) | HyperPCM (AUPRC/AUROC) | GraphDTA-style (AUPRC/AUROC) | Note | Winner |
|---|---|---|---|---|---|---|---|---|
| `BindingDB_Kd / unseen_drug` | 3 | 0.5509 ± 0.0900 / 0.7794 ± 0.0534 | **0.7339 ± 0.0051 / 0.8854 ± 0.0088** | 0.5004 ± 0.0341 / 0.7364 ± 0.0122 | 0.5694 ± 0.0473 / 0.7778 ± 0.0448 | 0.5626 ± 0.0172 / 0.7988 ± 0.0129 | adding a learner-control baseline rewrites the synthetic drug-cold ordering | RF |
| `BindingDB_Kd / blind_start` | 3 | 0.3737 ± 0.0229 / 0.6778 ± 0.0151 | **0.4512 ± 0.0373 / 0.7451 ± 0.0260** | 0.3636 ± 0.0693 / 0.6490 ± 0.0642 | 0.3828 ± 0.1024 / 0.6807 ± 0.0824 | – | the earlier "no reliable winner" row becomes RF-favored in the expanded panel | RF |
| `BindingDB_Kd / unseen_target` | 3 | **0.5224 ± 0.0602 / 0.7741 ± 0.0161** | 0.5183 ± 0.0612 / 0.7838 ± 0.0218 | 0.5136 ± 0.0730 / 0.7970 ± 0.0159 | 0.4955 ± 0.0793 / 0.7810 ± 0.0191 | 0.4722 ± 0.0925 / 0.7637 ± 0.0314 | synthetic target-cold reverts to `base` on mean AUPRC | base |
| `BindingDB_patent / patent_temporal` | 3 | 0.7772 ± 0.0086 / 0.7223 ± 0.0109 | **0.8234 ± 0.0012 / 0.7773 ± 0.0017** | 0.7825 ± 0.0078 / 0.7362 ± 0.0083 | 0.7753 ± 0.0050 / 0.7368 ± 0.0090 | 0.7726 ± 0.0016 / 0.7232 ± 0.0007 | temporal/provenance OOD changes again once the learner-control baseline is added | RF |
| `BindingDB_nonpatent_Kd / nonpatent_temporal` | 3 | 0.6369 ± 0.0173 / 0.7333 ± 0.0081 | **0.7201 ± 0.0012 / 0.7627 ± 0.0020** | 0.6531 ± 0.0066 / 0.7644 ± 0.0041 | – | 0.6635 ± 0.0142 / 0.7334 ± 0.0096 | the non-patent temporal probe confirms that the expanded panel is not LM-dominated | RF |

bootstrap interval for `RF - base` is strictly positive, and `RF` also remains above `HyperPCM` under paired bootstrap. On `blind_start`, the earlier "no reliable winner" result disappears once `RF` is added: `RF` reaches $0.4512 \pm 0.0373$, compared with $0.3737 \pm 0.0229$ for `base`, and the paired bootstrap interval for `RF - base` remains strictly positive. On `unseen_target`, however, the synthetic target-cold story is more stable. `base` remains strongest by mean AUPRC at $0.5224 \pm 0.0602$, while `RF` is close but slightly lower at $0.5183 \pm 0.0612$, and the paired bootstrap interval for `RF - base` crosses zero.

These results matter because they show that "synthetic cold-start DTI" is already not a single evaluation condition. Different synthetic regimes reward different model families, and the expanded panel shows that this is not merely a pooled-LM phenomenon. A classical learner on exactly the same handcrafted drug/target representation as `base` can become the strongest model on two synthetic axes, while the graph-family probe remains competitive but does not dominate. The review-driven similarity diagnosis sharpens this point further: on synthetic `unseen_drug`, the held-out compounds still have median nearest-train Tanimoto $0.7732$ and Murcko scaffold match rate $0.6204$. When we replace entity holdout with a stricter scaffold split, the earlier `HyperPCM` edge is no longer supported. Over three seeds on `BindingDB_Kd / scaffold_drug`, `base` reaches $0.6069 \pm 0.0161/0.8351 \pm 0.0037$, while `HyperPCM` reaches $0.5983 \pm 0.0368/0.8155 \pm 0.0204$, and the paired bootstrap interval for `HyperPCM - base` crosses zero (Appendix Tables A5 and A6). The synthetic panel therefore remains useful, but both its drug-cold ordering and its apparent best learner are sensitive to how chemically strict the split actually is.

## 4.2 Real temporal shift favors a different learner again

We next evaluate the expanded panel on the promoted temporal benchmarks. Once `RF` is added, the real-OOD picture changes again. On `BindingDB_patent / patent_temporal`, `RF` becomes the strongest model by mean AUPRC, reaching $0.8234 \pm 0.0012$, compared with $0.7772 \pm 0.0086$ for `base`, $0.7825 \pm 0.0078$ for `DTI-LM`, $0.7753 \pm 0.0050$ for `HyperPCM`, and $0.7726 \pm 0.0016$ for `GraphDTA-style`. Figure 1 therefore needs a stronger reading than in the earlier four-model draft: synthetic cold splits favor different systems, but the promoted temporal benchmarks now favor a classical learner-control baseline rather than a pooled-LM system. This advantage is statistically strong. Paired bootstrap gives `RF - base` $= +0.0463$ with 95% CI $[+0.0441, +0.0484]$ and `RF - DTI-LM` $= +0.0409$ with 95% CI $[+0.0384, +0.0433]$.

A second temporal robustness probe strengthens this reading further. On the non-patent benchmark built directly from the official BindingDB release, `RF` again becomes the strongest model over matched 3-seed evaluation, reaching $0.7201 \pm 0.0012/0.7627 \pm 0.0020$, versus $0.6531 \pm 0.0066/0.7644 \pm 0.0041$ for `DTI-LM`, $0.6369 \pm 0.0173/0.7333 \pm 0.0081$ for `base`, and $0.6635 \pm 0.0142/0.7334 \pm 0.0096$ for `GraphDTA-style`. Here too, the paired bootstrap intervals stay strictly positive for both `RF - base` and `RF - DTI-LM`. These additions change the scientific read of the temporal axis. The earlier conclusion that "pooled-LM features are best on temporal OOD" no longer survives the expanded panel. The stronger and more defensible conclusion is that temporal benchmarks do not preserve the ranking seen on synthetic panels, and that a classical learner on fixed handcrafted inputs can dominate both synthetic and temporal regimes.

This is the central revised result. The synthetic panel does not produce a single best model, and the promoted temporal benchmarks do not preserve the smaller panel's earlier ordering either. Instead, both benchmark axis and model family change the apparent ranking within the studied panel. The alternative-cutoff benchmark from the earlier revision remains useful because it still shows that temporal ordering is not an

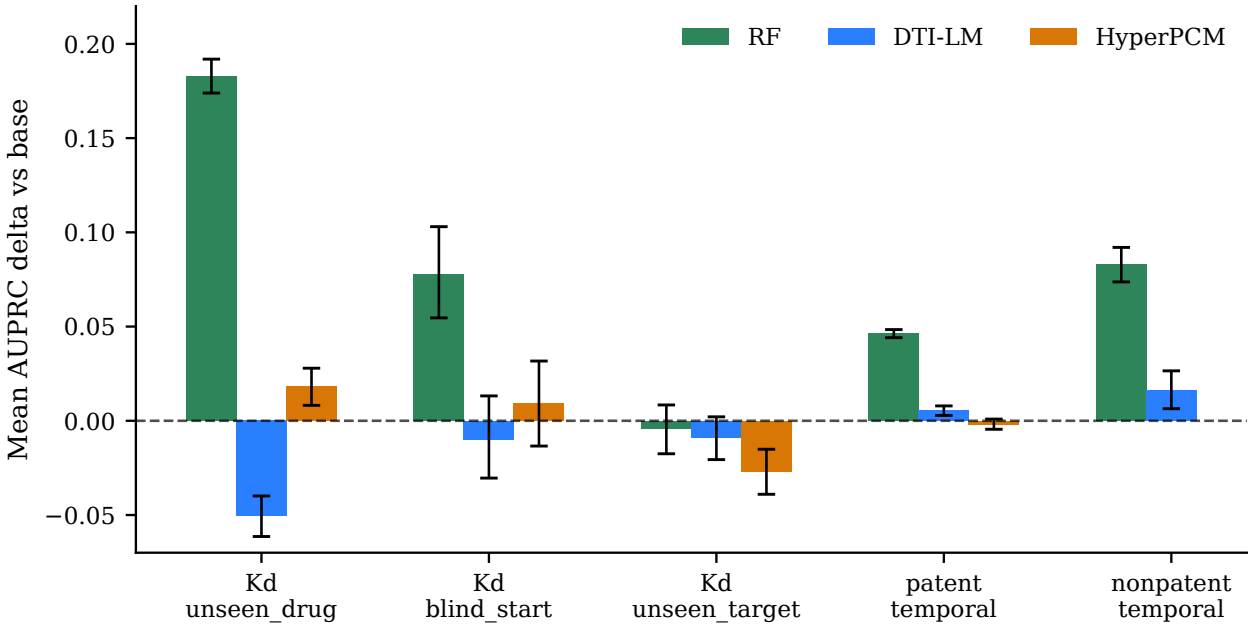


**Figure 1:** Mean within-benchmark AUPRC delta relative to `base` under the expanded panel. Bars are shown for the compared systems with paired-bootstrap support currently available. Because prevalence differs across benchmarks, the vertical scale should be read as a row-wise within-benchmark contrast rather than as a cross-benchmark effect-size scale.

artifact of a single year boundary, but the expanded five-model panel makes the broader point more strongly: the apparent best system can change not only across benchmarks but also across learner families built on the same underlying representation.

### 4.3 Temporal shift is structured across year bands and overlap levels

To help explain why the temporal/provenance benchmark changes the ranking, we analyze the patent results by year band and by overlap bucket. Table 3 makes the primary AUPRC slices explicit, while Figure 2 visualizes the same structure. To avoid over-reading the tiny 2021 cell, we pool 2020 and 2021 into a single recent stratum. Under the expanded panel, `RF` is strongest in both the large 2019 slice and the pooled 2020–2021 slice, so the year-band view no longer supports a simple "LM wins temporal OOD" interpretation. Instead, it shows that the aggregate temporal ranking is not tied to one model family alone.

The overlap-bucket analysis reveals another useful pattern. In the lowest-overlap bucket, `DTI-LM` still remains the strongest system, but `RF` becomes strongest in the middle- and high-overlap buckets and therefore dominates the aggregate temporal benchmark. The expanded panel thus strengthens, rather than weakens, the structured-shift diagnosis: temporal OOD is not just a colder version of synthetic cold-start, because different overlap strata continue to favor different model families. The benchmark outcome depends on both the shift axis and the modeling family, not only on whether a model uses pooled LM features.

### 4.4 Supporting target-cold OOD benchmarks still favor the plain base model

We treat `BindingDB_Ki / unseen_target` and `DAVIS / unseen_target` as supporting real-OOD benchmarks rather than primary headline axes. Under matched three-seed evaluation, neither promoted external baseline overturns the support-panel story. On `BindingDB_Ki`, `base` reaches $0.6934 \pm 0.0257/0.7192 \pm 0.0068$, compared with $0.6648 \pm 0.0177/0.6936 \pm 0.0044$ for `DTI-LM` and $0.6515 \pm 0.0301/0.6757 \pm 0.0211$ for `HyperPCM`. On `DAVIS`, `base` reaches $0.4097 \pm 0.0275/0.8613 \pm 0.0153$,

**Table 3:** Compact AUPRC summary behind the patent diagnosis under the expanded panel. Values are mean ± seed std over the matched 3-seed runs. The tiny 2021 slice is pooled into the `2020-2021` stratum to avoid over-interpreting a 72-example cell.

| Slice type | Slice | Count | base | RF | DTI-LM |
|---|---|---|---|---|---|
| Year | 2019 | 42,009 | $0.7626 \pm 0.0098$ | $\mathbf{0.8062 \pm 0.0013}$ | $0.7734 \pm 0.0071$ |
| Year | 2020–2021 | 7,019 | $0.8716 \pm 0.0121$ | $\mathbf{0.9098 \pm 0.0005}$ | $0.8533 \pm 0.0100$ |
| Overlap bucket | 0 (0.000–0.531) | 16,341 | $0.5551 \pm 0.0190$ | $0.6005 \pm 0.0034$ | $\mathbf{0.6353 \pm 0.0175}$ |
| Overlap bucket | 1 (0.531–0.710) | 16,342 | $0.7218 \pm 0.0167$ | $\mathbf{0.7606 \pm 0.0012}$ | $0.7489 \pm 0.0094$ |
| Overlap bucket | 2 (0.710–0.989) | 16,345 | $0.9237 \pm 0.0030$ | $\mathbf{0.9426 \pm 0.0005}$ | $0.8866 \pm 0.0097$ |

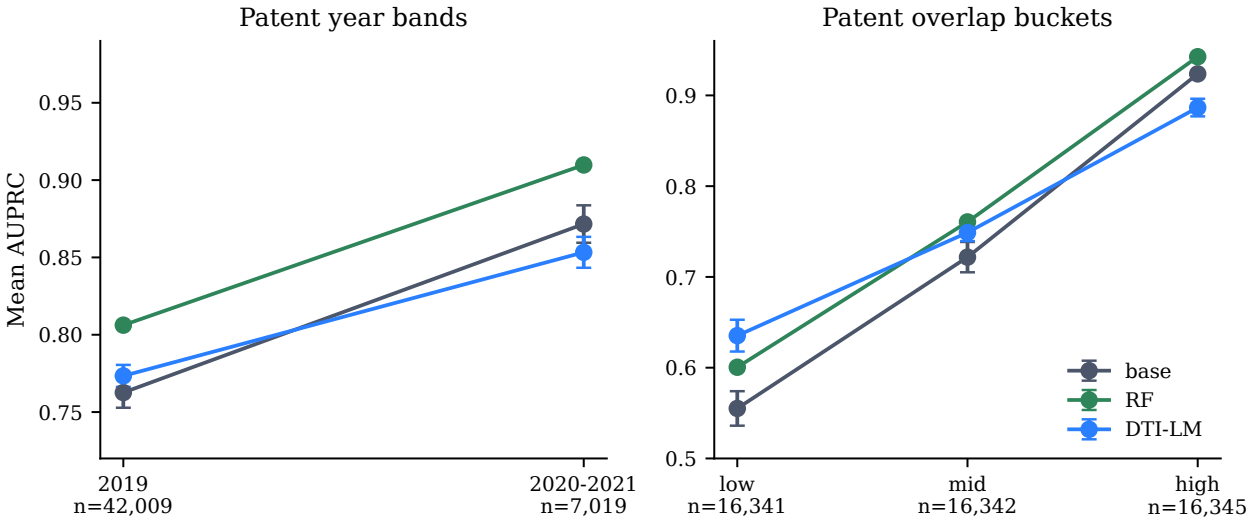


**Figure 2:** Patent temporal/provenance diagnosis in terms of mean AUPRC under the expanded panel. Left: pooled year-band slices, with 2021 merged into the recent stratum. Right: test-set overlap buckets computed from the saved overlap score. Error bars denote seed std over the matched 3-seed runs. In the expanded panel, `RF` dominates the aggregate temporal benchmark, while `DTI-LM` remains strongest only in the lowest-overlap bucket.

compared with $0.4090 \pm 0.0570/0.8718 \pm 0.0152$ for `DTI-LM` and $0.3840 \pm 0.0269/0.8610 \pm 0.0245$ for `HyperPCM`.

The `DAVIS` result is the closest supporting-panel case: `DTI-LM` is effectively tied with `base` on mean AUPRC and slightly higher on mean AUROC, but it still does not create a clean ordering separation of the kind seen on the promoted patent benchmark. `BindingDB_Ki` is more decisive, with `base` ahead of both external baselines on both metrics. These auxiliary results are therefore consistent with the main claim of the paper, but they do not carry the same argumentative weight as the patent benchmark because they do not add a cleaner shift axis than temporal/provenance OOD.

## 5 Discussion

The main message of this study is that, within the studied expanded panel, DTI evaluation is benchmark-sensitive in a way that current reporting practice often obscures. A synthetic split can still be useful, but it should not be treated as a universal proxy for real OOD generalization. Once the panel includes pooled-LM baselines, a graph-family probe, and a learner-control classical baseline on the same handcrafted inputs as `base`, the sign and magnitude of an apparent gain can still change when the benchmark changes.

An important design choice in this paper is the use of a deliberately expanded but still finite external model panel. A larger model zoo could be added later, but the present panel is already broad enough to invalidate the narrow objection that the earlier ranking changes were artifacts of comparing only pooled-LM variants. In particular, `RF` and `base` share the same handcrafted inputs but use different learners, while `GraphDTA-style` adds a graph-family inductive bias. The new results therefore support a stronger statement than in the smaller panel: benchmark-qualified ranking changes persist even when the comparison spans classical, fixed-feature neural, pooled-LM, and graph-based systems. At the same time, the present paper should still not be read as an exhaustive field-wide census.

The paper also has clear limitations. Although the support benchmarks `BindingDB_Ki` and `DAVIS` remain useful counterexamples, they are still supporting rather than co-equal headline axes. The expanded model family is much stronger than the original in-repo probe panel, but it is still partial rather than exhaustive. We now add two targeted robustness probes, and both make the manuscript more disciplined. The non-patent temporal benchmark shows that the temporal ordering is not confined to patent provenance alone, while the scaffold-drug probe shows that the original synthetic `unseen_drug` split was chemically permissive enough that the earlier `HyperPCM` edge does not survive a stricter holdout. Even with these additions, the patent benchmark is still temporal/provenance OOD rather than a pure time-only shift, so the current paper should not claim that time alone explains the observed ranking change. A further caveat is that ChemBERTa and ESM2 were pretrained on broad corpora that may already include molecular and protein entities appearing in our temporal test sets. We therefore cannot rule out that part of the earlier `DTI-LM` edge on temporal benchmarks reflects broader pretraining coverage of chemical or protein space rather than temporal robustness per se. At the same time, the expanded panel now shows that the main temporal winner is no longer a pooled-LM model but a classical learner-control baseline using the same handcrafted representation as `base`. This weakens any mechanistic story that rests only on LM pretraining and shifts the more credible conclusion toward benchmark sensitivity and learner sensitivity. We also do not evaluate downstream biological decision quality. This paper should therefore be read as a well-supported benchmark case study rather than an exhaustive census of all real-OOD DTI settings. These limitations should be stated plainly, but they do not weaken the more disciplined central finding that synthetic and real-OOD rankings can disagree sharply in the tested expanded panel.

# 6 Conclusion

We presented a benchmark-first study of distribution shift in DTI prediction. In the expanded five-model panel, a 3-seed synthetic reference panel on `BindingDB_Kd` showed that cold-start regimes are already heterogeneous: `RF` is strongest on synthetic `unseen_drug` and `blind_start`, while `base` remains strongest on synthetic `unseen_target` by mean AUPRC. Review-driven robustness checks then made the split semantics sharper: a scaffold-drug probe removed the earlier `HyperPCM` edge, and a non-patent temporal probe showed that the expanded temporal ordering is not confined to the patent benchmark alone. Under matched 3-seed evaluation on both `BindingDB_patent` and `BindingDB_nonpatent_Kd`, the strongest model is `RF`, while the `GraphDTA-style` graph-family probe remains competitive but not dominant. Pooled patent year-band and overlap-bucket analyses further showed that real OOD is structured rather than monolithic. Overall, these results indicate that, within the studied expanded panel and tested benchmarks, synthetic cold splits and real distribution shifts should not be treated as interchangeable evaluation settings, and that learner choice can matter as much as representation family. This manuscript should be read as a benchmark/resource case study rather than an exhaustive field-wide census. Future DTI papers should therefore report both synthetic and real-OOD evidence when making generalization claims.

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**Table A1:** Explicit specification of the promoted patent benchmarks. The temporal partition is fixed by year; the random seed changes model stochasticity, not the partition itself.

| Aspect | `patent_temporal` | `patent_temporal_v2017` |
|---|---|---|
| Source files | `train_val.csv + test.csv` | `train_val.csv + test.csv` |
| Train years | 2013–2017 | 2013–2016 |
| Validation year | 2018 | 2017 |
| Test years | 2019–2021 | 2018–2021 |
| Rows (train / valid / test) | 142,275 / 41,155 / 49,028 | 90,771 / 51,504 / 90,183 |
| Pos. rate (train / valid / test) | 0.566 / 0.534 / 0.575 | 0.568 / 0.562 / 0.556 |
| Label transform | raw patent value $Y \mapsto \mathrm{p}K_d = 9 - Y/\ln(10)$; positive label if $\mathrm{p}K_d \geq 7.0$ | |
| Overlap policy | no cold-entity filtering; entity overlap is allowed, exported, and diagnosed via `overlap_score` buckets | |
| Duplicate policy | pair rows are kept after year partition; entity export tables are deduplicated by canonical ID | |
| Seed semantics | partition is fixed across seeds; seed affects training / initialization only | |
| OOD interpretation | temporal/provenance OOD rather than pure time-only shift | |

**Table A2:** Explicit specification of the non-patent temporal probe. This benchmark is built from the official BindingDB TSV after filtering to non-patent, positive-$K_d$, single-chain rows with a valid publication year. The partition is fixed by publication year; the random seed changes model stochasticity, not the split.

| Aspect | `nonpatent_temporal` |
|---|---|
| Source file | official `BindingDB_All_202603_tsv.zip`, filtered to empty `Patent Number` |
| Date field | publication year from `Date of publication` |
| Train years | 1977–2018 |
| Validation year | 2019 |
| Test years | 2020–2023 |
| Rows (train / valid / test) | 35,755 / 3,006 / 7,174 |
| Pos. rate (train / valid / test) | 0.341 / 0.445 / 0.426 |
| Label transform | raw $K_d$ in nM with $\mathrm{p}K_d = -\log_{10}(Y/10^9)$; positive label if $\mathrm{p}K_d \geq 7.0$ |
| Overlap policy | no cold-entity filtering; `drug_overlap`, `target_overlap`, and `overlap_score` are exported for diagnosis |
| Duplicate policy | exact duplicate rows are removed before partitioning; no exact duplicates remain in the released split files |
| Seed semantics | partition is fixed across seeds; seed affects training / initialization only |
| OOD interpretation | non-patent temporal shift; cleaner than patent provenance, but still not a pure time-only intervention |

*Nature Machine Intelligence*, 7(3):495–508, 2025. doi: 10.1038/s42256-025-00987-y. URL `https://doi.org/10.1038/s42256-025-00987-y`.

# A Appendix

## A.1 Patent benchmark specification

Because the main paper depends on `BindingDB_patent / patent_temporal`, we make its construction explicit here. Table A1 enumerates the year boundaries, split counts, label construction, overlap policy, and seed semantics for both the main patent split and the earlier-cutoff robustness variant.

## A.2 Non-patent temporal probe specification

Because the main revision also relies on `BindingDB_nonpatent_Kd / nonpatent_temporal` to separate temporal drift from patent provenance, we make that construction explicit as well. Table A2 lists the source file, year cutoffs, label transform, prevalence, duplicate policy, and seed semantics for the released non-patent temporal split.

**Table A3:** Specification of the `base` reference model used throughout the benchmark. This table is included to make the anchor system reproducible rather than treating it as an informal baseline label.

| Component | Specification |
|---|---|
| Drug input | RDKit Morgan fingerprint, radius 2, 2048 bits, generated from canonical SMILES in `src/raicd/featurizers.py`. |
| Target input | L2-normalized hashed amino-acid 1/2/3-mer count vector, 2048 dimensions, generated in `src/raicd/featurizers.py`. |
| Drug encoder | Two-layer MLP: $\mathrm{Linear}(2048, 512)$, GELU, dropout $0.1$, $\mathrm{Linear}(512, 256)$, followed by L2 normalization. |
| Target encoder | Same architecture as the drug encoder, with independent parameters. |
| Pair head | Concatenate $[d,\ t,\ d \odot t,\ \lvert d - t \rvert] \in \mathbb{R}^{1024}$, then $\mathrm{Linear}(1024, 512)$, GELU, dropout $0.1$, $\mathrm{Linear}(512, 1)$. |
| Trainable parameters | 2,886,145 parameters in total. |
| Optimization | BCE-with-logits objective, Adam with learning rate $10^{-3}$ and weight decay $10^{-4}$, 8 epochs, batch size 256. |
| Labeling | Binary label derived from a `pKd` threshold of 7.0 (or patent-specific log-nM conversion for `BindingDB_patent`; see Table A1). |
| Code path | Model implementation in `src/raicd/model.py`; training entry point in `scripts/train_raicd.py`. |

**Table A4:** Secondary real-OOD support benchmarks under the expanded panel. These panels use matched 3-seed evaluation, but remain supporting evidence because they reinforce rather than replace the temporal benchmark story established in the main panel. To keep promotion focused on the decision-relevant synthetic and temporal axes, the support table reports the promoted external baselines rather than the full five-model panel.

| Benchmark | base (AUPRC/AUROC) | DTI-LM (AUPRC/AUROC) | HyperPCM (AUPRC/AUROC) | Winner |
|---|---|---|---|---|
| `BindingDB_Ki / unseen_target` | **0.6934 ± 0.0257 / 0.7192 ± 0.0068** | 0.6648 ± 0.0177 / 0.6936 ± 0.0044 | 0.6515 ± 0.0301 / 0.6757 ± 0.0211 | base |
| `DAVIS / unseen_target` | **0.4097 ± 0.0275 / 0.8613 ± 0.0153** | 0.4090 ± 0.0570 / 0.8718 ± 0.0152 | 0.3840 ± 0.0269 / 0.8610 ± 0.0245 | base |

## A.3 Reference-model specification

Because every ranking claim is anchored against `base`, we make that system explicit rather than treating it as an unnamed baseline. Table A3 specifies the exact input features, network structure, parameter count, optimization recipe, and implementation path used in every experiment.

## A.4 Additional real-OOD support benchmarks

The main paper focuses on the strongest evidence chain: the synthetic `BindingDB_Kd` reference panel, the promoted temporal benchmarks, and the patent year/overlap diagnosis. `BindingDB_Ki` and `DAVIS` remain supporting real-OOD benchmarks: both now use matched 3-seed evaluation, but they reinforce rather than replace the temporal story. To keep the promotion budget focused on the most decision-relevant axes, the support table keeps the promoted external baselines rather than rerunning the full five-model panel.

On the `BindingDB_Ki` support split, Table A4 shows that `base` remains ahead of both `DTI-LM` and `HyperPCM`. On `DAVIS`, `base` and `DTI-LM` are nearly tied on mean AUPRC, but `base` stays slightly higher on the paper's primary ranking metric. Together, these support panels reinforce the main message that strong recent baselines do not transfer uniformly across OOD definitions.

## A.5 Review-driven robustness probes

The external review highlighted two concrete risks: that the promoted patent result might be driven mainly by provenance rather than temporal shift, and that the synthetic `unseen_drug` split might still be chemically

**Table A5:** Targeted robustness probes added after the external review. The non-patent temporal probe tests whether the promoted patent ranking persists when patent provenance is removed. The scaffold probe tests whether the synthetic `unseen_drug` ordering survives a stricter chemistry-aware holdout. Values are mean $\pm$ std over three seeds.

| Benchmark | Systems | Mean AUPRC / AUROC | Paired AUPRC CI | Read |
|---|---|---|---|---|
| `BindingDB_nonpatent_Kd / nonpatent_temporal` | `base` vs `DTI-LM` | `base`: 0.6369 $\pm$ 0.0173 / 0.7333 $\pm$ 0.0081; `DTI-LM`: **0.6531 $\pm$ 0.0066 / 0.7644 $\pm$ 0.0041** | +0.0163 [+0.0064, +0.0265] | temporal ordering difference survives without patent provenance |
| `BindingDB_Kd / scaffold_drug` | `base` vs `HyperPCM` | `base`: **0.6069 $\pm$ 0.0161 / 0.8351 $\pm$ 0.0037**; `HyperPCM`: 0.5983 $\pm$ 0.0368 / 0.8155 $\pm$ 0.0204 | −0.0086 [−0.0203, +0.0025] | stricter scaffold holdout does not preserve the original `HyperPCM` edge |

**Table A6:** Paired bootstrap support for the expanded-panel comparisons. Positive deltas favor the second model in the comparison; negative deltas favor the first.

| Comparison | Mean AUPRC delta | 95% CI |
|---|---|---|
| `unseen_drug`: `RF - base` | +0.1830 | [+0.1739, +0.1919] |
| `unseen_drug`: `RF - HyperPCM` | +0.1645 | [+0.1558, +0.1733] |
| `blind_start`: `RF - base` | +0.0775 | [+0.0546, +0.1030] |
| `scaffold_drug`: `HyperPCM - base` | −0.0086 | [−0.0203, +0.0025] |
| `unseen_target`: `RF - base` | −0.0042 | [−0.0175, +0.0084] |
| `patent`: `RF - base` | +0.0463 | [+0.0441, +0.0484] |
| `patent`: `RF - DTI-LM` | +0.0409 | [+0.0384, +0.0433] |
| `nonpatent_temporal`: `DTI-LM - base` | +0.0163 | [+0.0064, +0.0265] |
| `nonpatent_temporal`: `RF - base` | +0.0832 | [+0.0737, +0.0920] |
| `nonpatent_temporal`: `RF - DTI-LM` | +0.0670 | [+0.0564, +0.0764] |

too permissive. Table A5 records the targeted fixes. The first probe constructs a non-patent temporal benchmark directly from the official BindingDB release and shows that even before the learner-control expansion, `DTI-LM` stays ahead of `base` over matched 3-seed evaluation. The second probe replaces entity holdout with a Murcko-scaffold holdout and shows that the original synthetic `HyperPCM` edge does not survive this stricter split.

## A.6 Paired uncertainty support

Table A6 summarizes the paired bootstrap intervals for the main external-panel comparisons against `base`. The most important headline claims keep the same sign under paired resampling: `HyperPCM` is reliably stronger than `base` on synthetic `unseen_drug`, and `DTI-LM` is reliably stronger than `base` on the promoted patent temporal benchmark. The remaining main-panel comparisons either stay negative or cross zero, which is exactly why the paper frames them as split-specific rather than universal advantages.

## A.7 Recent baseline screening and promotion

To avoid cherry-picking concerns, Table A7 summarizes the seed0 screening pass used to decide which recent baselines to promote. The promotion criterion was simple: prioritize the strongest recent model on the primary real-OOD benchmark `patent_temporal`, then keep any additional recent baseline that is competitive enough to test whether the main conclusion changes.

## A.8 Reusable benchmark resources

The benchmark release is intended to be reused independently of the in-repo models. Table A8 lists the concrete artifacts exported for each split and how they interact with the standalone evaluator. In practice,

**Table A7:** Transparent seed0 screening used to decide which recent baselines were promoted to matched three-seed evaluation. The primary promotion criterion was performance on `BindingDB_patent / patent_temporal`, the main real-OOD benchmark.

| Model | `patent_temporal` | `blind_start` | `unseen_target` | Promotion | Rationale |
|---|---|---|---|---|---|
| `DTI-LM` | 0.7897 / 0.7435 | 0.3873 / 0.7130 | 0.5308 / 0.8011 | main 3-seed promotion | strongest seed0 recent baseline on the primary real-OOD benchmark; therefore promoted in the main text |
| `HyperPCM` | 0.7686 / 0.7247 | 0.3896 / 0.7447 | 0.5294 / 0.7990 | secondary 3-seed promotion | competitive on synthetic seed0 axes, but weaker on the main patent benchmark; promoted only to test whether the main conclusion changes |

**Table A8:** Reusable benchmark artifacts released with each split. The release is designed so that external predictions can be scored without re-running the in-repo training code.

| Artifact | Example files | Purpose |
|---|---|---|
| Split manifest | `manifest.json` | versioned metadata, split counts, prevalence, seed, and canonical paths |
| Partition tables | `train.csv`, `valid.csv`, `test.csv` | canonical labeled partitions with `example_id` and exported overlap scores |
| Pair tables | `train_pairs.csv`, `test_pairs.csv` | lightweight pair-only views for external model pipelines |
| Entity tables | `drugs.csv`, `targets.csv` | unique drug / target vocabularies aligned to the exported split IDs |
| Full joined table | `all.csv`, `all_pairs.csv` | complete resource snapshot for auditing or custom slicing |
| Standalone evaluator | `scripts/evaluate_prediction_csv.py` | scores a prediction CSV against the canonical reference split and returns global + overlap-bucket metrics |

an external user only needs the canonical split bundle plus a prediction CSV keyed by `example_id`; the evaluator then reproduces the same global metrics and overlap-bucket metrics reported in this paper.

## A.9 Method-centric ablations

Earlier project stages explored routers, conservative target-memory variants, and selective fallback rules. These analyses were useful for method diagnosis, but they do not define the final paper narrative. The benchmark paper is intentionally centered on ranking instability across shift definitions rather than on proposing a new best-performing method.